INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

Aircraft Fuel Cell Ventilation System

The General Services Administration has authorized the use of this Commercial Item Description for all federal agencies.

1. SCOPE.

1.1 Scope. This Commercial Item Description (CID) covers the minimum design requirements for a transportable Aircraft Fuel Cell Ventilation System (FCVS).

2. SALIENT CHARACTERISTICS.

2.1 Fuel cell ventilation system description. The aircraft fuel cell ventilation system shall allow work crews to transport and set-up an aircraft fuel systems repair area. The contractor shall provide a van body trailer stowed mobile aircraft fuel tank ventilation and personnel respiratory protection system for use during aircraft fuel systems maintenance in an explosive environment.

2.1.1 Towable storage enclosed trailer. All of the provided system components shall be transported and stowed in an enclosed weather proof, highway towable, and air transportable trailer. The trailer interior shall be equipped with a storage rack consisting of separate storage compartments accommodating the individual components to prevent any damage during transport and also provide additional space for tools, pylons, and other minor gear associated with the operation. Trailer design shall conform to the requirements of Department of Transportation (DOT) commercial standards for highway use and the MIL-STD-1791 requirements for air transportability. Maximum exterior dimensions and weight shall be adequate to meet the storage requirements but not to exceed maximums in 2.1.1.1. Trailer running gear shall be at a minimum rated to the actual Gross Vehicle Weight Rating (GVWR) determined in accordance with 4.5.5.

2.1.1.1 Weight and dimensions.

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2.1.1.1.1 Maximum weight and dimensions. Overall weight of trailer, including all system components, and external dimensions shall not exceed:

Weight	3000 pounds.
Length	176 inches.
Width	102 inches.
Height	94 inches.

The maximum dimensions given above are based on using two 463L air transport pallets joined together with a coupler.

2.1.1.1.2 Objective Trailer dimensions. A desired objective is to have the trailer fit on a single 463L pallet. The maximum allowable dimensions, in transport configuration, to fit on a single pallet are:

Length	102 inches (see 2.1.1.7)
Width	84 inches.
Height	94 inches.

2.1.1.2 Trailer body design. The construction shall be either trussed or monocoque, single or double axle, and leak proof meeting the environmental requirements of 2.3. Vertical members (corner and intermediate posts) shall be on maximum of 12” centers between roof structure and chassis frame/floor structure, and be continuous without splice. Longitudinal members (upper rails at the roof and underbody) shall be continuous without splice. Roof shall be crowned to permit drainage. Flooring cross members shall be plated, powder coated, or primed and painted to preclude corrosion. A minimum ¾” thick treated lumber floor decking shall be used. The decking shall be coated with a rubber paint. An undercoating shall be applied following commercial practice. The trailer shall use a drop down rear door with cable lift assist to serve as ramp and equipped with a lock. The door shall be full width, equipped with a continuous hinge. The interior of the door shall be coated with rubber paint. The height of the door shall be 72” minimum and equipped with weatherproof gaskets and drip molding.

2.1.1.3 Trailer electrical. Light emitting diode (LED) type lighting shall be used for 12 Volts Direct Current (VDC) tail lights, brake lights, clearance lights, and turn signal. A J560b receptacle w/spring loaded cover shall be located on the front, centerline of the trailer. A J560b lug/harness securable to tow bar and of sufficient length to connect the trailer to the tow vehicle allowing clearance for turning shall be provided. A 120 Volts Alternating Current (VAC) interior lighting system shall be used to provide full illumination for personnel working inside trailer. Powered by external source and equipped with a circuit breaker protected equipment base receptacle, mounted on the exterior of the trailer and recessed with a spring loaded cover. An electric cord, minimum length of 25 feet shall be provided to connect the trailer lighting to a power source.

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2.1.1.4 Tires and wheels. Tire and wheel assemblies shall be identical at all wheel positions. Tires and wheels shall be in accordance with the *Tire and Rim Association Year Book* requirements for this application. The wheels shall be constructed with aluminum alloy.

2.1.1.5 Spare tire and wheel. The trailer shall be equipped with a spare tire and wheel identical to 2.1.1.4.

2.1.1.6 Tire pressure. Tire pressure shall not exceed 100 pounds per square inch (psi) and shall not be reduced for air transport.

2.1.1.7 Tow bar. A retractable tow bar may be used to reduce the stored trailer dimensions to the objective length (see 2.1.1.1.2).

2.1.1.8 Coupling. The trailer shall be equipped with a solid lunette measuring 3.00 inch inside diameter and 1.625 inch section diameter rated at least the rated capacity of the trailer. The coupling will be capable of being towed by vehicles with a pintle height of 14 to 25 inches.

2.1.1.9 Safety chains. Two heavy duty safety chains shall be provided.

2.1.1.10 Parking brakes. The trailer shall have manually applied, internally expanding, parking brakes. The control handle shall be mounted on the front of the trailer and have provisions for slack adjustment.

2.1.1.11 Service brakes. The trailer shall have hydraulic or electric surge type service brakes. Hydraulic fluid shall be in accordance with MIL-PRF-83282. If electric brakes are used a brake controller shall be provided.

2.1.1.12 Fenders. The trailer shall have steel fenders in accordance with best commercial practice.

2.1.1.13 Axles. The trailer shall be equipped with a single or twin axle(s) rated at least equal to the rated capacity of the trailer. Wheel base is not to exceed maximum width given in 2.1.1.1.

2.1.1.14 Suspension. The trailer shall be equipped with a spring suspension system with components having a rated capacity at least equal to the GVWR.

2.1.1.15 Tie downs. The trailer shall be symmetrically restrained during air and ground transport. A minimum of 4 tie down points shall be rated at a minimum of 5,000 pounds each, marked for capacity, with a clear opening compatible with MIL-DTL-25959 and MIL-PRF-27260 tie down devices. Each end of each tie down device shall terminate at a tie down point and not pass through any other tie down point. There shall be no interference between tie down devices and the trailer. The tie down provisions shall be in accordance with 4.1 through 4.12 of MIL-STD-209.

2.1.1.16 Trailer jack. A double wheel trailer jack shall be rated 6.5 times load at trailer rated GVWR in accordance with MIL-STD-1791.

2.1.1.17 Stabilizing jacks. The trailer shall be equipped with stabilizing jacks.

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2.1.1.18 Landing pad. A landing pad for the trailer jack wheel in accordance with MIL-STD-1791 is required. A storage area for the landing pad shall be provided in the trailer.

2.1.1.19 Fork lift provisions. The trailer shall be equipped with fork lift pockets.

2.1.1.20 Storage rack. The trailer shall contain compartmented storage racks designed to secure the individual items which comprise the FCVS. The storage compartments shall be labelled in accordance with 2.2.2. The items shall be secured in accordance with the transportability requirements in 2.5.

2.1.2 Aircraft fuel cell ventilation system components. The FCVS shall contain electrically driven, explosion proof and intrinsically safe components that meet or exceed National Fire Protection Code (NFPA) 70, Class I Division 1 hazardous area standards. The system shall be divided into two separate sub-systems; one which shall provide for the ventilation of aircraft fuel tanks to an acceptable entry safe Lower Explosion Limit (LEL), the second shall simultaneously provide full-faced, supplied air respiratory protection for a minimum of three fuel systems maintenance personnel.

2.1.2.1 Electrically driven breathing air pump. The breathing air unit shall be a portable, electrically driven oil-less pump. It shall provide continuously filtered ambient air for a minimum of three personnel simultaneously while wearing National Institute for Occupational Safety and Health (NIOSH) approved full faced supplied air respirators. The unit shall be explosion proof and meet NFPA 70, Class I Division 1 hazardous area standards. The unit must operate on 115 volt, single phase, 60 Hz electrical current, with no more than 20 amp draw. The unit shall have a minimum electric cord length of 25 feet. The unit must draw clean air from a remote source using a snorkel that has a replaceable filter element designed to remove particulate matter one-half micron or larger. The unit shall deliver 15 CFM, at 10 psi, to operate three NIOSH approved supplied air full face respirators simultaneously. The unit shall have rubber wheels and be equipped with a minimum of two lifting handles. Any surface of the breathing unit that comes in contact with the ground must have a rubber style foot as to prevent the unit from coming in direct contact with the surface. There shall be provided one electrically driven breathing air pump equivalent to Rhine Air P/N NF23-1.

2.1.2.2 Breathing air pump filter cartridge kit. A kit of replacement disposable filter cartridges shall be provided for the breathing air pump. The filter cartridge shall be made of borosilicate glass microfibers and have 1/8” wall thickness. A case (10 each) shall be provided equivalent to Rhine Air P/N CF8080.

2.1.2.3 Breathing air pump air inlet extension. An air inlet extension shall be provided to allow the breathing air unit intake to be relocated without moving the breathing air unit itself. It shall be a minimum of 5 feet high at the intake, have a minimum hose length of 50 feet and be made of a flexible, UV resistant material. It shall have a cam style locking feature that allows for easy attachment to the breathing air unit without modification. There shall be one breathing air pump inlet extension provided equivalent to Rhine Air P/N IFX-5500.

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2.1.2.4 Respirators. A full faced, supplied air, Type C, respirator that meet OSHA standard 29 CFR 1910.134 and current NIOSH certifications, is required. The respirator shall work in concert with the supplied air breathing pump and maintain their NIOSH certification when operating. There shall be a 6-point adjustable head strap and provisions to attach two organic vapor filter cartridges on each side of the respirator and a low pressure regulator with a quick disconnect connector. The respirator hose shall attach to the supplied breathing air hose via a belt mounted low pressure fitting. Three respirators shall be provided equivalent to 3M Full Face Reusable Respirators 7800 series P/N 7800S-S (small) P/N 7800S-M (Medium) P/N 7800S-L (Large) Low Press Regulator Kit P/N W-3197.

2.1.2.5 Respirator hoses. Respirator hoses, 1/2" x 100', that meet OSHA standard 29 CFR 1910.134 and current NIOSH certifications capable of suppling 4 CFM of breathable air to each respirator from the shall be provided. The hoses shall have a quick disconnect feature and attach to the breathing air pump at one end and the low pressure fitting with belt at the other end. Three respirator hoses shall be provided equivalent to Rhine Air P/N W-3020-100.

2.1.2.6 Respirator filters. Two filter cartridges shall be mounted on the exterior of the respirators and serve as a backup source of filtered air if the primary supplied air system fails. The respirator filters shall be of the organic vapor type and be made of sorbent activated carbon. Filter cartridges shall meet OSHA standard 29 CFR 1910.134 and current NIOSH certifications. There shall be one case (60 each) disposable organic vapor filters per kit equivalent to Rhine Air P/N 70070316776.

2.1.2.7 Electric 12" air ventilation unit. Two portable electric driven explosion proof 12" centrifugal blowers shall be provided. The units shall have the ability to perform both a positive and negative purge of an aircraft fuel tank. The units shall be explosion proof and meet NFPA 70, Class I Division 1 hazardous area standards. The blower duct interfaces shall have 'J Slot' quick disconnect style system and work in conjunction with the 12 inch air ventilation ducting described in 2.1.2.8. The blower shall have a minimum discharge flowrate of 1500 CFM at 5 inch water static pressure. The unit must operate on 115 volt, single phase, 60 Hz electrical current, with no more than 20 amp draw. The unit shall have a minimum electric cord length of 25 feet. The unit shall be equipped with a minimum of two lifting handles. Any surface of the ventilation unit that comes in contact with the ground must have a rubber style foot as to prevent the unit from coming in direct contact with the surface. The electric air ventilation units supplied shall be similar to Rhine Air P/N AF-178E NOTE: This part number is identified as an 8" outlet/inlet and will require modification to adopt to the 12 inch duct.

2.1.2.8 Air ventilation conductive ducting. There shall be 12" x 15' long individual neoprene coated 2 ply polyester fabric hoses that provide a means to dissipate static electricity when bonded together and bonded to a grounding point. The hoses shall have a spring steel helical construction and have a 'J Slot' stainless steel style quick disconnect system on the ends, male on one end and female on the other to permit chaining the ducts together. Electrical continuity between ends shall not exceed 10 ohms. Each end shall be fabricated from stainless steel integrally grounded to the duct work and designed to mate directly to the ring sections of the fan exhaust or discharge openings. Both end sections shall be stamped with: "WARNING Ground duct before using". There shall be ten ducts supplied with kit equivalent to NSN 4720009305974.

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2.1.2.9 Bonding cables. Bonding cables shall be 50 foot in length and made of 3/32-inch diameter seven strand stainless steel cables and have a maximum allowable resistance of 10 OHMS. They shall be coated in polyester elastomer yellow insulation. The following cables shall be supplied; three each MIL-DTL-83413/7-1 clamp -to- MIL-DTL-83413/7-1 clamp, three each MIL-DTL-83413/4-1 bayonet -to- MIL-DTL-83413/4-1 bayonet and three each MIL-DTL-83413/4-1 bayonet-to- MIL-DTL-83413/7-1 clamp.

2.1.2.10 Extension cord for hazardous locations. Electrical extension cord assemblies, 50' in length and rated at 20 amps, designed to work in conjunction with the breathing air pump and the air ventilation systems. The extension cords shall be explosion proof and meet NFPA 70, Class I Division 1 hazardous area standards. There shall be three extension cords provided with kit equivalent to Rhine Air P/N EXC-50.

2.2 Design and construction. The design shall promote cost effective, life-cycle sustainability by addressing considerations such as incorporating open standards, reducing pollutant emissions and wastes, and increasing fuel economy, while satisfying system performance requirements. It shall be designed and constructed so that no parts will work loose in service, and to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. It shall be weatherproof and designed to prevent the intrusion of water, sand, and dust into critical operating components.

2.2.1 Materials, protective coatings, and finish.

2.2.1.1 Protective coatings. Materials that deteriorate when exposed to sunlight, weather, or operational conditions normally encountered during the service life of the item shall not be used or shall have means of protection against such deterioration that does not prevent compliance with the performance requirements specified herein. Protective coatings that chip, crack, or scale with age or extremes of climatic conditions or when exposed to heat shall not be used. Exposed surfaces of fasteners, handles, and fittings shall also be primed and painted. See 2.2.4.1 for prohibited hazardous materials.

2.2.1.2 Finish. Unless otherwise specified (see 6.2), the exterior finish color of the trailer shall be grey, color number 26173 grey in accordance with SAE AMS-STD-595. The inner surfaces of the trailer compartments shall be finished using commercially available color. FCVS components shall be coated using the best commercial practice. All aluminum or steel surfaces shall be optionally finished color number 26173, grey in accordance with SAE AMES-STD-595 using epoxy primer and two part polyurethane top coat or powder coating systems. Anodized aluminum surfaces are acceptable. Unpainted stainless steel surfaces are acceptable.

2.2.1.3 Exclusion of water. The design of the trailer shall be such as to prevent water leaking into, or being driven into, any part of the trailer interior when either in an operating or travelling configuration. All doors, panels, covers, etc., shall be provided with sealing arrangements such that the entry of water is minimized when these items are correctly closed. Particular care shall be taken to prevent wetting of equipment. Sharp corners and recesses shall be avoided so that moisture and solid matter cannot accumulate to initiate localized attack. Sealed floors with suitable drainage

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shall be provided for storage compartments and other areas in the trailer that could collect and retain water.

2.2.1.3.1 Fluid traps and faying surfaces. There shall be no fluid traps on the trailer or system components. Faying surfaces of all structural joints, except welded joints, shall be sealed to preclude fluid intrusion.

2.2.1.3.2 Ventilation. Ventilation shall be sufficient to prevent moisture retention and buildup.

2.2.1.3.3 Drainage. Drain holes shall be provided to prevent collection or entrapment of water or other unwanted fluid in areas where exclusion is impractical. All designs shall include considerations for the prevention of water or fluid entrapment and ensure that drain holes are located to effect maximum drainage of accumulated fluids. The number and location of drain holes shall be sufficient to permit drainage of all fluids when the trailer is on a 10-degree longitudinal slope facing both up and down or side slope in each direction (right and left side facing up the slope). The minimum size of the drain holes shall be 0.375 inch.

2.2.2 Markings. All external devices which require an operational or maintenance interface shall be marked in accordance with MIL-STD-130. Markings shall be applied with decals and shall be 1-inch high block letters unless prohibited by the available space. In such cases, the markings shall be the largest size possible. Markings, Information/Caution shall be Lusterless Black, Color Number 37038 of SAE AMS-STD-595, and Markings, Warning/Danger shall be Lusterless Red, Color Number 31136 of SAE AMS-STD-595. The center of gravity of the trailer shall be marked on the unit within 1-inch of the calculated center of gravity. Tire inflation pressure shall be marked as near to each tire as practicable.

2.2.3 Identification and information plates.

2.2.3.1 Identification plate. An identification plate in accordance with MIL-STD-130 shall be securely attached to the trailer in a readily accessible location. The identification plate shall contain the following information: nomenclature, part number, serial number, date of manufacture, manufacturer's name, Commercial and Government Entity (CAGE) code, date of warranty expiration, and National Stock Number (NSN). The FCVS components for which the Government's unit cost is more than $5,000, is serially managed, or the procuring agency determines is mission essential, shall have Unique Identification (UID) (also known as Item Unique Identification (IUID)) information permanently affixed on or near the respective identification plate(s), marked in accordance with MIL-STD-130. UID information shall be included as both a bar code and human readable markings.

2.2.3.2 Transportation data plate. A transportation data plate shall be securely attached to the trailer in a readily accessible location. The plate shall contain at least the following information:

a. Side and rear silhouette views of the trailer.

b. Horizontal and vertical location of the center of gravity of the trailer in air transportable configuration, marked on the silhouette views.

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c. Shipping weight.

d. Loading cubage.

e. Overall height, width, and length.

f. Jack stand and axle loads.

g. Tie down information.

2.2.3.3 Lubrication data plate. The trailer and FCVS components having lubrication requirements shall be provided with a permanently marked lubrication data plate constructed of a non-corroding metal, mounted on the exterior surface that shall direct attention to all lubrication fittings and components that require lubrication. The plate shall identify the type and grade of lubricant required for operational temperatures specified in 2.3.1.

2.2.4 Environment, Safety, and Occupational Health (ESOH).

2.2.4.1 Hazardous material. The design shall minimize and control hazards associated with the inclusion or use of hazardous or toxic materials and the generation of toxic or noxious gases. The FCVS shall not generate or use Class I or Class II Ozone Depleting Substances (ODS) during operation, maintenance, or disposal. Class I ODS and hazardous materials shall not be used in any system, component, or process. The FCVS shall not contain or use either hexavalent chromium or cadmium without written approval by the procuring activity. Hazardous materials are defined in AFI 32-7086; Class I and Class II ODS are defined in Title 40 of the Code of Federal Regulations (40CFR82).

2.2.4.2 Component protection. All space in which work is performed during operation, service, and maintenance shall be free of hazardous protrusions, sharp edges, or other features which may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to personnel, shall be guarded or insulated to eliminate the hazard. All wires, cables, tubes, and hoses shall be supported and protected to minimize chafing and abrasion and shall be located so as to provide adequate clearance from moving parts and high operational temperatures. Grommets shall be provided wherever wires, cables, tubes, or hoses pass through bulkheads, partitions, or structural members.

2.2.4.3 Foreign object damage (FOD). All loose metal parts, such as pins or connector covers, shall be securely attached to the FCVS with wire ropes or chains. "Dog tag" style beaded chains shall not be provided. Removable panels, if provided, shall be attached with captive fasteners. Tire valve stem caps shall be made of plastic.

2.2.4.4 Electrostatic discharge (ESD). The design of the FCVS shall preclude equipment damage due to ESD, protect personnel from electrical shock due to static charging, and prevent ignition of explosive atmospheres due to sparking.

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2.2.5 Fastening devices. All screws, bolts, nuts, pins, and other fastening devices shall be properly designed, manufactured, and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers, or nylon patches shall not be used for this purpose, except for the attachment of trim items or as provided in commercial (see 6.3.1) components. Tapped threads shall have a minimum thread engagement in accordance with Table I.

TABLE I. Minimum thread engagement.

Material	Minimum Thread Engagement
Steel	1.0 times the nominal fastener diameter
Cast iron, brass, or bronze	1.5 times the nominal fastener diameter
Aluminum, zinc, or plastic	2.0 times the nominal fastener diameter

2.2.6 Welders and welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof, and service loadings.

2.3 Environmental conditions.

2.3.1 Operating temperature range. The FCVS shall be capable of operating in ambient temperatures ranging from -25° F to 125° F.

2.3.2 Storage temperature range. The FCVS shall be capable of being stored in ambient temperatures ranging from -40° F to 140° F.

2.3.3 Precipitation.

2.3.3.1 Rain. The trailer shall be capable of storage during rainfall of 5-inches per hour for three consecutive hours and 10-inches per hour for 10 consecutive minutes, with winds of up to 35 knots; and with 6-inches of rain per hour impinging on the trailer at angles from vertical to 45° for 30 consecutive minutes.

2.3.3.2 Snow. The trailer shall be capable of storage during accretion of wet snow up to 2-inches per hour for at least 12 hours.

2.3.3.3 Ice. The trailer shall be capable of storage with ice accretion up to 1.5-inches on exposed horizontal surfaces.

2.3.4 Solar radiation. The trailer shall not be adversely affected by full time exposure to solar radiation, such as those conditions encountered in desert environments.

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2.3.5 Fungus. All materials used in the FCVS shall be fungus resistant or shall be suitably treated to resist fungus. Materials treated for fungus resistance shall retain their original electronic and physical properties, shall not present toxic hazards, and treatment shall last for the entire service life of the part. The FCVS shall be suitable for operation and storage in conditions encountered in a tropical environment.

2.3.6 Salt fog. The FCVS shall be capable of storage and operation in high temperature, high humidity, salt laden, and sea coast environments without damage or deterioration of performance.

2.3.7 Sand and dust. The trailer shall be capable of storage during exposure to wind-blown sand or dust without damage or deterioration of performance.

2.5 Transportability.

2.5.1 Surface transportability. The trailer shall be transportable via a suitable tow vehicle for on highway, improved and unimproved roads (see 6.3.3 and 6.3.5), and the air base ramp environment.

2.5.2 Air transportability. The trailer shall be transportable on C5, C17, and C130 aircraft in accordance with MIL-STD-1791. All components of the FCVS shall be secured inside the trailer with the exception of the trailer jack landing pad.

2.5.2.1 Equipment removal and reconfiguration. The normal storage configuration of the FCVS components inside the trailer shall be in accordance with 2.5.2 and not requiring reconfiguration for air transportation.

2.5.2.2 Flight loads. The FCVS shall be capable of being restrained and withstanding, without loss of serviceability, 2.0 G up and 4.5 G down accelerations, and shall be capable of being restrained and withstanding, without loss of structural integrity, 3.0 G forward, 1.5 G aft, and 1.5 G lateral accelerations.

2.5.2.3 Trailer venting. The trailer shall be equipped with pressure relief devices or configured for air transport to prevent any part from becoming a projectile in the event of catastrophic loss of aircraft cabin pressure.

2.5.2.4 Aircraft loading of the trailer. The trailer shall be capable of being secured to a single 463L pallet or two 463L pallet(s) which have been coupled together.

2.6 Maintenance.

2.6.1 Inspection and servicing provisions.

a. Pre-use inspections and servicing tasks shall not require tools.

b. Routine service and preventive maintenance shall not require special tools (see 6.3.3).

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c. The FCVS shall be designed with maximum usage of sealed lifetime lubricated bearings.

2.6.2 Special tools. The design of the item shall minimize the requirement for special tools (see 6.3.4). All special tools shall be provided with, and stored in the trailer.

2.7 Workmanship. The FCVS, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Workmanship objectives shall include freedom from blemishes, defects, burrs and sharp corners and edges; accuracy of dimensions, surface finish, and radii of fillets; thoroughness of welding, painting, and riveting; marking of parts and assemblies; and proper alignment of parts and tightness of assembly fasteners.

2.7.1 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be deburred. Threaded fasteners shall be tight and shall not work loose during testing or service usage.

2.7.2 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be deburred. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the component.

2.7.3 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustment, gear assemblies shall be free of excessive backlash.

2.7.4 Cleaning. The FCVS shall be thoroughly cleaned. Loose, spattered, or excess solder; welding slag; stray bolts, nuts, and washers; rust; metal particles; pipe compound; and other foreign matter shall be removed during and after final assembly.

3. REGULATORY REQUIREMENTS.

3.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

3.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

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4. PRODUCT CONFORMANCE PROVISIONS.

The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace, modified as necessary to comply with the requirements herein. The Government reserves the right to require proof of such conformance.

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Pre-production inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Pre-production inspection. The pre-production FCVS shall be subjected to the analyses, demonstrations, examinations, and tests described in 4.5.1 through 4.5.7. The contractor shall provide or arrange for all test equipment and facilities, except for the compatibility test described in 4.5.7.

4.3 Conformance inspection. Each production FCVS shall be subjected to the examination described in 4.5.1.

4.4 Inspection requirements.

4.4.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory precision type, calibrated at proper intervals to ensure laboratory accuracy.

4.4.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

d. Ambient temperature.

e. Ambient humidity.

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4.4.3 Test rejection criteria. Throughout all tests specified herein, the FCVS shall be closely observed for the following conditions, which shall be cause for rejection.

a. Failure to conform to design or performance requirements specified herein.

b. Any spillage or leakage of any liquid, including lubricant, or hydraulic fluid, under any condition, except as allowed herein.

c. Thermal or structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear. If excessive wear is suspected, the original equipment manufacturer (OEM) specifications or tolerances shall be utilized for making a determination.

e. Evidence of corrosion or deterioration.

f. Misalignment of components.

g. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

h. Interference between the FCVS components or between the trailer, the ground, and all required obstacles, with the exception of normal contact by the tires.

i. Evidence of undesirable mobility characteristics, including instability in handling during cornering, braking, and while traversing all required terrain.

4.5 Test methods.

4.5.1 Examination of product. Each FCVS shall be examined to verify compliance with the requirements herein prior to accomplishing any other demonstrations or tests listed in 4.5. A contractor-generated, Government-approved checklist (part of the test procedure) shall be used to identify each requirement not verified by an analysis, certification, demonstration, or test, and shall be used to document the examination results. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and sealants and their application, welding, fastening, and markings. Proper operation of each FCVS function shall be verified. Certifications and analyses shall be provided in accordance with Table II. Each production FCVS shall be inspected to a Government-approved reduced version of the checklist.

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TABLE II. Certifications and analyses.

Paragraph	Required Certifications and Analyses
2.1.1 Towable storage enclosed trailer.	Certification stating trailer running gear are rated at or above GVWR.
2.1.1.15 Tie downs.	Contractor tie down provision analysis (see 4.5.6.2.2).
2.1.1.20 Storage rack.	Contractor storage rack analysis (see 4.5.6.2.1)
2.1.2 Aircraft fuel tank ventilation and personnel respiratory protection system components	Certification stating the electrical components met or exceed NFPA 70, Class I Division 1 hazardous standards.
2.1.2.4 Respirators.	Certification stating the full faced, supplied air, Type C, respirators meet OSHA standard 29 CFR 1910.134 and current NIOSH certifications.
2.2.4.4 Electrostatic discharge (ESD).	Contractor analysis of the grounding and bonding system(see 4.5.2.6)
2.1.2.5 Respirator hoses.	Certification stating respirator hoses meet OSHA standard 29 CFR 1910.134 and current NIOSH certifications.
2.3.3.2 Snow.	Contractor analysis of the snow load requirement (see 4.5.3.3.2).
2.3.4 Solar radiation.	Contractor certification stating the FCVS performance is not adversely affected by full time exposure to solar radiation, such as those conditions encountered in desert environments.
2.3.5 Fungus.	Contractor certification that the materials used in construction of the FCVS are fungus resistant or suitably treated to resist fungus.
2.5.2 Air transportability.	Contractor air transportability analysis (see 4.5.6.2).

4.5.2 Verifications. The pre-production FCVS shall be subjected to the following tests and analysis.

4.5.2.1 Blower performance test. Electric current draw and air flow output at the blower discharge shall be in accordance with 2.1.2.7 operated over a continuous 24 hour period.

4.5.2.2 Breathing air compressor performance test. Electric current draw and airflow output at the compressor discharge shall be in accordance with 2.1.2.1 operated over a 24 hour period. The inlet extension and filter shall be installed for this test.

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4.5.2.3 Trailer functionality. Each item stored within the trailer shall be tested for proper fit, component lock-in-place function, and labelling. All trailer doors, locks, lights and jacks will be tested for freedom of operation and weatherproof/environmental function capability.

4.5.2.4 Air ventilation conductive duct sections. Each duct section shall be tested for conductivity, frictional loss, dimensional and material conformance. Each section shall be subjected to the maximum static pressure and CFM for the blower or fan it serves for a period of 24 hours with no degrading of material or performance.

4.5.2.5 Respirator hoses. All hoses and connectors shall be verified for conformance to the material specifications and withstand pressure testing for a period of 24 hours at 200% of their maximum rated capacity as specified herein before with no greater than 3% loss in capacity.

4.5.2.6 Electrostatic discharge analysis. Contractor analysis of the grounding and bonding system confirming compliance with 2.2.4.4.

4.5.3 Environmental testing.

4.5.3.1 High temperature storage and operation test. A pre-production FCVS shall be tested in accordance with MIL-STD-810, High Temperature Test Method, Procedures I and II, to demonstrate compliance with the high temperature storage and operating requirements of 2.3.1 and 2.3.2. Test duration shall be one 24-hour cycle for each procedure beginning no less than two hours after test item temperature stabilization.

4.5.3.2 Low temperature storage and operation test. A pre-production FCVS shall be tested in accordance with MIL-STD-810, Low Temperature Test Method, Procedures I and II, to demonstrate compliance with the low temperature storage and operating requirements of 2.3.1 and 2.3.2, Test duration shall be one 24-hour cycle for each procedure beginning no less than two hours after test item temperature stabilization.

4.5.3.3 Precipitation.

4.5.3.3.1 Rain test. A pre-production shall be tested in accordance with MIL-STD-810, Rain Test Method, Procedure I, to demonstrate compliance with 2.3.3.1.

4.5.3.3.2 Snow load analysis. An engineering analysis shall be performed to demonstrate compliance with the snow load requirement of 2.3.3.2, using a specific gravity of snow of 0.1 (Ref. 5.3 of MIL-STD-810, Part Three).

4.5.3.3.3 Ice accretion test. A pre-production trailer shall be tested in accordance with MIL-STD-810, Icing/Freezing Rain Test Method with an ice thickness of 1.5-inches to demonstrate compliance with the ice accretion requirement of 2.3.3.3.

4.5.3.3.4 Salt fog test. A pre-production FCVS shall be tested in accordance with MIL-STD-810, Salt Fog Test Method, to demonstrate compliance with 2.3.6. Test duration shall be alternating

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24-hour periods of salt fog exposure and drying conditions for 24-hour periods (two wet and two dry).

4.5.3.3.5 Sand and dust test. A pre-production trailer shall be tested in accordance with MIL-STD-810, Sand and Dust Test Method, Procedures I (12 hours) and II (90 minutes per side), to demonstrate compliance with 2.3.7.

4.5.5 Weight and dimension verification.

4.5.5.1 Weight and center of gravity test. The weight, center of gravity, and trailer jack weight of the pre-production trailer shall be measured to demonstrate compliance with the weight requirement of 2.1.1.1 and the center of gravity marking requirement of 2.2.2.

4.5.5.2 Dimension measurement. A pre-production trailer shall be measured to demonstrate compliance with the dimensional requirements of 2.1.1.1.

4.5.6 Transportability verification.

4.5.6.1 Surface transportability verification.

4.5.6.1.1 Surface transportability road test. Surface transportability shall be verified for the pre-production FCVS by road testing the trailer loaded with all system components. Using a suitable tow vehicle;

a) Maintain an average speed of 25 miles per hour on unimproved roads for at least 50 miles. No system components shall break loose from the stored location or be damaged. No excessive swaying of the trailer when control of the vehicle is questioned is permitted. Braking action shall be smooth with no lock up of the trailer wheels.

b) Maintain an average speed of 60 miles per hour on a dry level paved road for at least 100 mile. No system components shall break loose from the stored location or be damaged. No excessive swaying of the trailer when control of the vehicle is questioned is permitted. Braking action shall be smooth with no lock up of the trailer wheels.

4.5.6.1.2 Parking brake test. The parking brake shall hold the trailer motionless on a 20 degree longitudinal grade in both upgrade and downgrade directions. The grade surface shall be dry, hard, and free from loose material. The force required to set the brakes under these conditions shall not exceed 55-pounds.

4.5.6.1.3 Service brake test. The trailer service brakes shall be tested during the 4.5.6.1.1 road testing. The braking action shall be smooth with no wheel lock up.

4.5.6.2 Air transportability analysis. An engineering analysis shall be performed to demonstrate compliance with the air transportability requirements of 2.5.2, 2.5.2.1, 2.5.2.2, 2.5.2.3 and 2.5.2.4. The analysis shall include the tie downs and all major components and their ability to withstand

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the accelerations specified in 2.5.2.2. The analysis shall demonstrate the accessibility and structural adequacy of the tie downs specified in 2.1.1.15.

4.5.6.2.1 Storage rack provision verification. The contractor shall perform an engineering analysis to demonstrate compliance with the rack requirements of 2.1.1.20.

4.5.6.2.2 Tie down provision analysis. An engineering analysis shall be performed to demonstrate compliance with the tie down provision requirements of 2.1.1.15.

4.5.7 Compatibility test. A user compatibility test will be conducted on a production unit in an operational environment which will task the unit to mission roles. The testing will be conducted by Air Force personnel at a location to be determined in the contract, after completion of the pre-production test. Any changes made to the pre-production article unit due to deficiencies found during the pre-production testing shall be made to the compatibility test article before shipping to the compatibility test location.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 Department of Defense and Federal documents. Department of Defense and Federal documents are available online at http://quicksearch.dla.mil/ or from the DLA Document Services, 700 Robbins Avenue, Building 4D, Philadelphia PA 19111-5094.

6.1.2 AFI documents. Copies of publication are available online at http://www.e-publishing.af.mil

6.1.3 ASTM documents. Application for copies should be addressed to ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken PA 19428-2959, http://www.astm.org/.

6.1.4 AWS documents. Application for copies should be addressed to American Welding Society, 550 N.W. LeJeune Road, Miami FL 33126, http://www.aws.org/.

6.1.5 CFR references. The Code of Federal Regulation (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402 or online at http://www.gpo.gov/fdsys/search/home.action .

6.1.5 NEMA documents. Application for copies should be addressed to NEMA, 1300 North 17th Street, Suite 1847, Rosslyn VA 22209, www.nema.org.

6.1.6 NFPA documents. Applications for copies should be addressed to National fire Protection Association, 1 Batterymarch Park, Quincy, MA 02169, http://www.nfpa.org .

6.1.7 NIOSH respirators. Contact the National Institute for Occupational Safety and Health, 1600 Clifton Road, Atlanta, GA 30329-4027. Phone 800-232-4636, https://www.cdc.gov/niosh/contact

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6.1.8 OSHA standards. OSHA standards may be obtained online at http://www.osha.gov/ or from Occupational Safety and Health Association, 200 Constitution Ave., Washington, D.C. 20210.

6.1.9 Rhine Air references. Informational references can be obtained at Rhine Air Inc., 10744 Prospect Ave, # A, Santee, CA 92071 or online at http://www.rhineair.com .

6.1.10 SAE documents. Applications for copies should be addressed to SAE International, 400 Commonwealth Drive, Warrendale, PA 15096, http://standards.sae.org/amsstd595/ .

6.1.11 Tire and Rim Association documents. Application for copies should be addressed to The Tire and Rim Association, Inc., 175 Montrose West Ave., Suite 150, Copley OH 44321, http://www.us-tra.org/.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this CID.

b. Packaging requirements (see 5).

6.3 Definitions.

6.3.1 Commercial item.

(1) Any item, other than real property, that is of a type customarily used by the general public or by non-Governmental entities for purposes other than Governmental purposes, and—

(i) Has been sold, leased, or licensed to the general public; or

(ii) Has been offered for sale, lease, or license to the general public;

(2) Any item that evolved from an item described in paragraph (1) of this definition through advances in technology or performance and that is not yet available in the commercial marketplace, but will be available in the commercial marketplace in time to satisfy the delivery requirements under a Government solicitation;

(3) Any item that would satisfy a criterion expressed in paragraphs (1) or (2) of this definition, but for—

(i) Modifications of a type customarily available in the commercial marketplace; or

(ii) Minor modifications of a type not customarily available in the commercial marketplace made to meet Federal Government requirements. Minor modifications means modifications that do not significantly alter the non-Governmental function or essential physical characteristics of an item or component, or change the purpose

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of a process. Factors to be considered in determining whether a modification is minor include the value and size of the modification and the comparative value and size of the final product. Dollar values and percentages may be used as guideposts, but are not conclusive evidence that a modification is minor;

(4) Any combination of items meeting the requirements of paragraphs (1), (2), (3), or (5) of this definition that are of a type customarily combined and sold in combination to the general public;

(5) Installation services, maintenance services, repair services, training services, and other services if—

(i) Such services are procured for support of an item referred to in paragraph (1), (2), (3), or (4) of this definition, regardless of whether such services are provided by the same source or at the same time as the item; and

(ii) The source of such services provides similar services contemporaneously to the general public under terms and conditions similar to those offered to the Federal Government;

(6) Services of a type offered and sold competitively in substantial quantities in the commercial marketplace based on established catalog or market prices for specific tasks performed or specific outcomes to be achieved and under standard commercial terms and conditions. For purposes of these services—

(i) “Catalog price” means a price included in a catalog, price list, schedule, or other form that is regularly maintained by the manufacturer or vendor, is either published or otherwise available for inspection by customers, and states prices at which sales are currently, or were last, made to a significant number of buyers constituting the general public; and

(ii) “Market prices” means current prices that are established in the course of ordinary trade between buyers and sellers free to bargain and that can be substantiated through competition or from sources independent of the offerors.

(7) Any item, combination of items, or service referred to in paragraphs (1) through (6) of this definition, notwithstanding the fact that the item, combination of items, or service is transferred between or among separate divisions, subsidiaries, or affiliates of a contractor; or

(8) A non-developmental item, if the procuring agency determines the item was developed exclusively at private expense and sold in substantial quantities, on a competitive basis, to multiple State and local Government. (Reference the Federal Acquisition Regulation (FAR) 2.101)

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6.3.2 Common hand tool. A non-powered tool that is likely to be found in a typical mechanic's toolbox. Common hand tools include open end, boxed end, combination, socket (both 6- and 12-point in both standard and deep-well), and hex key wrenches, in SAE sizes up to and including 1-inch and metric sizes up to and including 25-mm; ratchet handles, extensions, and swivels; slotted and Phillips-head screwdrivers; regular and snap-ring pliers; and a ball-peen hammer.

6.3.3 Improved road. An improved road is a smooth, hard surfaced road such as a concrete or asphalt paved highway.

6.3.4 Special tool. A tool that is not commercially and readily available from a source other than the FCVS contractor.

6.3.5 Unimproved road. An unimproved road is an unsurfaced road of dirt, gravel, or similar construction.

6.4 Key words.

Borosilicate glass microfibers
FCVS (Fuel Cell Ventilation System)
Transportable trailer
Van body trailer

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